

Empirical Proof Record: VALIDATED

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Proof ID: c1c889feafde5139a20f0ecf7ca35f0f...

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1. Claim

On the live substrate, growing the geoid dimension mid-session via `_refit_geoid_dimension` (re-fit on the substrate's own lived cloud, grow-only, $N=IPR$) keeps the cognitive cycle coherent — post-grow mean $\Phi \geq 0.5$ — where a naive dimension swap collapses it ($\Phi=0$, stale Ymir + F2 drift). N grows with accumulated experience.

- **Operationalisation:** Learned- N Takwin at $N=16 \rightarrow$ warm-up cycles on real BGE words $\rightarrow$ `_refit_geoid_dimension()` ($N=\text{round } IPR$ of the lived `_geoid_map`) $\rightarrow$ post-grow cycles; observed = post-grow mean Φ , gated on `grew=True`. (Measured in the Kimera venv; signed from the cache.)
- **Threshold:** `post_grow_mean_phi >= 0.5 phi`
- **H0:** H0: post-grow mean $\Phi < 0.5$ or N did not grow — a live mid-session grow degrades cognition
- **H1:** H1: post-grow mean $\Phi \geq 0.50$ and N grew — runtime dynamic- N is viable

2. Verdict

VALIDATED — observed 0.638

live mid-session grow $N 16 \rightarrow 71$ ($IPR 71.0$, 152 geoids re-projected, `grew=True`); post-grow mean $\Phi 0.638$ over 5 cycles (pre-grow 0.652); naive-swap baseline $\Phi=0$ (stale Ymir + F2 drift; journals 077/079) — the method's resets keep the cycle coherent

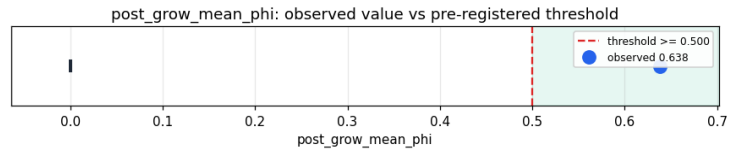


Figure 1: `post_grow_mean_phi` – observed value with Wilson 95% CI.

3. Evidence

Pillar	Statistic	Value	95% CI	Library
<code>runtime_dynamic_n_coherent</code>	<code>post_grow_mean_phi</code>	0.638	(0.0000, 0.0000)	numpy 2.4.4

4. Pre-registration

- Registered at: 2026-05-24T17:57:17.824255+00:00
- Config hash: c84c34eae94682a0f9c02e4bae6eaea4...
- Data hash: 8193bfba90344687c19a747691d10b4b...

5. Signature

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